DNA and genetic engineering

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Cloning, genetic engineering, DNA ... what is all the fuss about?

Find out how to extract your own DNA at home!

If someone gave you some salt, flour, citric acid, icing sugar, milk powder, baking soda, and fruit drink powder from the kitchen, would you know how to make Sherbet? Simply mixing all these ingredients together will produce nothing but a sticky mess!

You can't make something useful without a recipe, just like you can't build a living thing without a plan.

DNA - The recipe for life

Back to our lesson...think of a whole organism such as yourself...what makes you human? You, a human, can be thought of as being made up of smaller pieces...a number of different organs.

Some of your organs are the brain, skin, the liver, kidneys, muscles, etc So what are organs made up of? Specialised cells. Muscles are made up of muscle cells, the skin is made up of skin cells, and so on. How does a skin cell know how to be a skin cell? It has a set of instructions of how to be one, instructions stored on chromosomes.

A human will never really have kittens because we don't have the instructions of how to make cat cells! A seed knows what type of plant it should look like because it has the set of instructions to make all the organs (leaves, roots, etc) it requires to look like the parent plant.

The plan is stored on thread-like chromosomes, made of a chemical called DNA. We can get the DNA plan out of cells and study it, change it or swap it!

To isolate plant DNA

- Blend 50g of Cauliflower in 200 mL of water for 30 seconds to get single cells.
- Strain through muslin cloth. Pour 15 mL of liquid into a screw cap tube with 3 drops of detergent and shake gently for 5 seconds.
- Carefully layer to top of tube with cold alcohol (or meths)*. Use a hooked glass rod to spool out strands of DNA at the interface.

To isolate your own human DNA

(an original protocol independantly developed by Nexus. Please acknowledge our contribution if using this protocol for teaching purposes)

- To a 1/2 cup of water dissolve about 1/2 teaspoon of salt and add a squirt of dish-washing detergent. Save this for step 3
- Swirl about 25 mL of water around your mouth for 30 seconds. This removes some cheek cells. Spit into a disposable cup
- Add about 2cm of the fluid to a test-tube (or a Fuji film cannister) and add about 1cm of the saline/ detergent solution. Invert gently 3 or 4 times to mix well (but you don't want a lot of froth). This will break open the many cheek cells you spat out, releasing the DNA message that each cell must carry.
- Layer on top some ice cold ethanol (or methanol)*. Strands of DNA will be seen where the two layers meet. Hook out the strands of DNA that form with a glass hook (or one made from a plastic twist-tie) by slowly dipping up and down through the two layers.

Chemical tests confirm the presence of DNA:-

The following chemical tests can be used to check you actually have DNA:

- Test for purines: Add excess 2M ammonia solution and a few drops of 0.1M silver nitrate to 1 mL of DNA extract. A white precipitate indicates the presence of purines.
- Test for phosphate: Add 1 mL of 0.2M ammonium molybdate (39.02g/L) to 0.5 mL of extracted DNA and warm gently at 60-70oC. DO NOT BOIL. Yellow colour indicates presence of phosphate.
- Test for deoxyribose: Add 2 mL of Disches reagent to 1 mL of extracted DNA. Boil in water bath for 15 minutes. Green-blue colour indicates presence of deoxyribose. *Disches reagent:* 486 mL glacial acetic acid, 14 mL conc. sulphuric acid, 5g diphenylamine. Stir well and add 500 mL distilled water

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* **SAFETY**: Please use appropriate safe handling techniques when dealing with chemicals or biological material

Genetic Engineering:-

Swapping Recipes:

DeoxyriboNucleic Acid (DNA) is the chemical message cells read to know what to do. Change the message and you change the type of cell....change the cell and you change the organ...change the organ and you change the whole organism. If the DNA from a luminescent jellyfish is inserted into the DNA plan of a mouse it could glow in the dark. This has actually been done! This creature is now both jellyfish and mouse...a totally new lifeform through genetic engineering.

DNA is made up of a series of only four compounds; Adenine, Guanine, Thymine and Cytosine; an example is: CAGATTCCAAACCGGGGGTTC; a unique piece of DNA "code" to tell the cell what to do. The code is read in sets of three:- CAG ATT CCA AAC CGG GGG TTC with each 3 letter combination telling the cell which one of the amino acids to use in assembling PROTEINS the cell uses for work (enzymes), its structure or shape, communication, movement, colour, etc. This simple set of four bases read in groups of three allows for an almost infinite number of different proteins to be made; an infinite number of different cells. All bacteria, viruses, fungi, plants and animals use this same code. We can swap DNA from one lifeform to another and it will be read!

Nature's genetic engineers - viruses:

Viruses do not follow the MRSGREN characteristics of living things. Scientists say that a living thing must do 7 things: move, reproduce, sense the environment, grow in size, respire, excrete wastes and use nutrients. Viruses do not sense the environment, do not grow in size, do not carry out respiration, do not make wastes, and do not consume food. A virus is simply a piece of DNA (or RNA - Ribonucleic Acid) that is wrapped in a protein coat. This is called the virus particle. Remember the points above about MRSGREN. We say viruses are living in the loosest sense because they can reproduce: A particle attaches itself to or enters a living cell. The DNA or RNA is released into the cell The cell can't tell the difference between its original DNA instructions and the new ones. It follows the new ones. New virus particles are assembled by the cell until it is so full (about 100 to 1000) that it bursts and dies. Each new particle will infect other cells.

Any drug that is used to stop viruses reproducing must stop the host cell reproducing - even the healthy ones! That is why there has never been a drug that can cure viral diseases. Viruses have very short pieces of DNA or RNA (about 1000 to 9000 bases compared to billions in plants or animals). Any change to the DNA or RNA (a mutation) caused by sunlight, chemicals, or copying errors in cells have a significant impact on how the virus behaves. It is estimated in a 24 hour period one strain of virus can mutate into 300 - 3000 different new strains. It may be possible that two different viruses may infect the same host without causing symptoms and form a hybrid eg, each year we see new strains of Flu virus that seem to be part bird virus and part human virus.

Website article at
www.nexusresearchgroup.com/fun_science/dna.htm

Electrophoresis

From Nexus Research Group – FUN Science and home experiments...

This is a technique that allows you to separate a mixture of molecules using an electric current. Fragments of DNA separated to form a "fingerprint" is one application of this technique.

Materials: Soap dish; 9 volt batteries; Aluminium foil OR nichrome wire; Agarose OR Agar; Baking Soda; Ice-cream container lid; Glycerol; Droppers or Pasteur pipettes; Test mixture (could be plant indicator material that is concentrated by evaporation or human cheek cell DNA); Methylene Blue (only if DNA was your test mixture).

BUFFER: Dissolve about 1/4 teaspoon Baking Soda in $1^1/_4$ cups water (3 g in 300ml water). In the lab you can make a 1% solution by dissolving 1g in 100ml water.

Gel: Agarose is traditionally used but may be expensive. Make a 1% agarose gel by putting about 1/4 teaspoon of agarose in $^1/_4$ cup buffer (1g in 100ml). Alternatively use agar (not nutrient agar) in its place. I have often used a 0.8% agar gel (0.8g in 100ml).

Gelcomb: Use a thin plastic lid similar to an ice-cream tub lid to cut a "comb" that will rest on the sides of the dish but have rectangular "teeth" with square ends pointing downward. The 5mm wide teeth need to be cut to a length that will stop 2 or 3mm from the bottom of the soap dish. See if you can get four or more teeth on a comb.

Electrophoresis chamber: Use a soap dish lid or its base.

1. Microwave the agarose or agar gel solution on High until all has dissolved (this may take about 5 minutes). Allow to cool to about 50°C (able to be held in hand without burning). If it gets too cool and starts to set, simply reheat! You can do this as many times as required.
2. Pour your "hand hot" gel solution into your dish. You need a gel between 5 and 7mm thick, no more. Immediately place you comb into the gel about 2 cm from one end. This will form "wells" in which to place your sample mixtures.
3. Leave to set on a level surface. Carefully remove the comb. We now want to prevent the gel from being distorted by the bubbles that form at the electrodes. Use a blade to cut out a 1cm wide strip of gel from opposite ends of the dish where the electrodes will go. Don't damage the wells that should now be 1cm from the new edge.
4. At each end, place a wide strip of aluminium foil covering the interior sides to act as electrodes, or alternatively, place a length of nichrome wire that overlaps one edge of the dish for attachment of an alligator clip.
5. **Hint**: To speed things up you could pour several gels and store in the fridge with plastic wrap covering the top. They will keep for quite a few days.

Loading the wells:

1. Flood the gel with buffer (ice cold sometimes helps) so that the gel is covered to a depth of about 3mm.
2. Mix 2 drops of glycerol with 6 drops of your sample on a slide. The glycerol will make your sample more dense so it will fall into the well rather than diffuse into the buffer and be "lost". Colourless samples can have a drop of Bromophenol Blue added.

3 Collect some of your sample in a dropper or pasteur pipette and place the tip just under the surface of the buffer above a well. Slowly fill the well. Coloured samples of plant material are easy to monitor.
4 Fill the other wells in a similar manner with the rest of your test samples. Don't forget to keep a record of what sample was in which well!

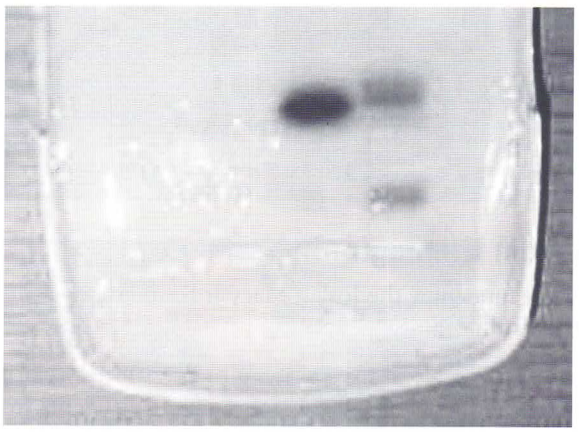
Running the gel:

1 Use an alligator clip to connect the electrode closest to the wells to the negative terminal of one of the 9 volt batteries. Connect 4 other batteries in series by stacking and connecting oppositely charged terminals in a pyramid fashion. The remaining positve terminal is connected to the other electrode with another alligator clip.
2 After a while you should see bubbles forming at the electrodes. Leave the gel until your samples have seperated into distinct bands.If you had to add Bromophenol Blue wait till it has almost run to the end of the gel.
3 If you have used plant or animal DNA in your sample, stain the gel with Methylene Blue (0.02 g in 100ml water). Stain for one hour (at least) then rinse in water.

Record your observations:

Points to note:

- NEVER use Ethidium Bromide to stain DNA. It is carcinogenic.
- A 1% gel with a slightly alkaline buffer around pH 8 works well.
- Sodium chloride buffer works but often causes the gel to heat up and melt.
- Platinum electrode wire can be obtained from Crescendo Supplies. We were fortunate when last year they generously donated some material for these trials.
- Use thin gels with small volume sample wells to get sharper bands.



Two different samples of food colouring run at 45 volts for 30 minutes

- Place the comb in the middle of the gel if comparing other dyes in the lab or other test material. Some bands may migrate to the positve terminal, others to the negative. You need to allow room for the differences to be seen!
- Molecules are often electrically charged. A molecule that has an overall positive charge is attracted to the negative electrode and vice-versa. DNA has an overall negative charge.
- Small molecules or DNA fragments tend to travel faster than large molecules or DNA fragments.

References:

Robert J Lebowitz.1995. *Plant Biotechnology: A laboratory manual.*
Charles Carter. 1998. Personal communication

Website article at
www.nexusresearchgroup.com/fun_science/electrophoresis.htm

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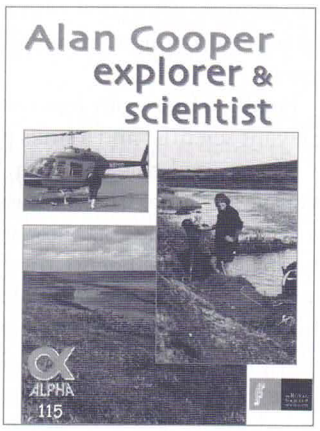

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